

SIOUX FALLS 14 NNE, SD, USA

WMO: 726600

Lat: 43.7346N

Lon: 96.6222W

Elev: 486

StdP: 95.62

Time Zone: -6.00 (NAC)

Period: 02-19

WBAN: 04990

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.1	-21.7	-27.4	0.3	-24.1	-25.0	0.4	-21.6	8.7	-7.9	7.8	-6.9	2.1	N/A	0.489

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	32.0	23.1	30.4	22.3	28.9	21.3	25.2	29.7	24.0	28.8	23.0	27.5	3.3	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.9	19.9	28.2	22.7	18.4	27.0	21.5	17.2	26.0	80.0	29.6	75.0	29.0	70.8	27.6	29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.3	6.4	5.6	-28.9	35.1	3.2	1.6	-31.2	36.2	-33.1	37.1	-34.9	38.0	-37.3	39.2
			-29.1	26.8	3.3	1.1	-31.5	27.6	-33.4	28.2	-35.2	28.8	-37.6	29.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.5	-9.0	-7.7	0.6	7.7	14.3	19.9	22.3	20.6	16.8	8.8	1.0	-6.5
	DBStd	12.34	7.23	6.85	7.24	5.86	4.93	3.48	3.07	2.96	4.39	5.40	5.93	6.45
	HDD10.0	2387	588	496	299	108	17	0	0	0	4	89	274	511
	HDD18.3	4298	847	729	549	319	144	23	5	12	79	299	521	769
	CDD10.0	1468	0	0	9	41	149	297	381	329	208	51	3	0
	CDD18.3	338	0	0	1	2	18	71	128	83	33	3	0	0
	CDH23.3	3116	0	0	5	42	231	649	1182	650	316	41	0	0
	CDH26.7	948	0	0	1	12	69	192	407	175	84	8	0	0
Wind	WSAvg	2.4	2.7	2.7	2.9	3.1	2.9	2.3	1.8	1.7	2.0	2.4	2.4	2.3
Precipitation	PrecAvg	695	14	20	37	73	94	112	84	69	57	31	20	
	PrecMax	943	35	57	104	172	202	327	210	199	186	156	109	53
	PrecMin	444	1	4	4	12	25	25	11	27	6	8	1	3
	PrecStd	126	8	12	25	36	47	56	55	47	46	43	30	14
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.6	12.8	22.9	28.7	31.9	33.4	35.2	32.8	31.5	27.8	19.8	10.1
		MCWB	4.2	8.9	12.5	16.5	18.2	22.9	24.2	24.2	22.0	16.2	12.5	6.3
	2%	DB	4.3	6.8	18.0	23.6	28.5	30.7	32.5	30.4	29.2	23.6	15.5	6.4
		MCWB	1.8	3.7	11.7	13.2	17.9	20.9	24.5	23.6	21.0	14.6	9.4	3.3
	5%	DB	2.3	3.8	14.1	20.4	25.8	28.9	30.6	28.6	27.0	20.5	12.1	3.4
		MCWB	0.9	1.9	10.0	11.8	16.8	20.5	23.6	22.3	19.3	13.1	7.5	1.6
	10%	DB	0.7	1.6	10.7	17.5	23.1	27.0	28.9	26.9	24.8	17.5	9.4	1.7
		MCWB	-0.3	0.2	7.5	10.2	15.6	19.6	22.3	21.2	18.6	11.4	6.1	0.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.5	9.8	15.9	16.4	21.1	25.5	27.2	26.1	23.8	18.0	12.4	7.9
		MCDB	8.3	13.6	20.6	23.7	26.4	31.2	31.8	30.6	28.9	24.3	17.5	9.9
	2%	WB	2.1	4.6	12.9	14.6	19.5	23.7	25.7	24.6	22.4	15.6	9.9	3.6
		MCDB	3.7	7.1	17.0	20.9	26.6	29.3	30.2	28.9	27.1	20.9	14.1	6.1
	5%	WB	1.2	2.3	10.5	12.5	18.2	22.4	24.6	23.4	21.4	14.1	7.8	2.0
		MCDB	2.5	4.2	15.0	17.6	24.4	27.4	29.2	27.4	25.2	18.7	11.6	3.8
	10%	WB	0.0	0.5	7.6	10.8	16.9	21.0	23.5	22.2	20.0	12.4	5.8	0.7
		MCDB	1.1	1.8	11.6	15.4	21.8	26.1	28.1	26.0	23.4	16.7	9.0	1.9
Mean Daily Temperature Range	5% DB	MDBR	9.1	9.5	10.1	11.8	11.7	11.0	11.5	11.1	12.2	11.3	10.6	8.7
		MCDBR	11.1	12.2	15.9	17.6	15.6	13.3	13.1	12.8	14.5	15.7	14.8	11.6
		MCWBR	9.1	9.2	10.0	9.4	7.2	6.5	6.0	6.1	6.9	8.4	9.2	8.7
	5% WB	MCDBR	10.2	12.0	16.0	15.0	13.1	12.0	11.7	11.5	12.1	13.9	13.5	11.2
		MCWBR	8.5	9.0	9.9	9.0	6.6	6.8	6.3	6.4	6.8	8.0	8.7	8.8
Clear-Sky Solar Irradiance	taub	0.269	0.282	0.316	0.368	0.384	0.394	0.400	0.403	0.355	0.324	0.291	0.272	
	taud	2.432	2.416	2.409	2.291	2.324	2.352	2.333	2.299	2.413	2.473	2.482	2.439	
	Ebn at Noon	873	921	925	896	888	877	866	850	871	858	838	832	
	Edh at Noon	73	88	102	125	125	122	123	123	101	83	69	66	
All-Sky Solar Radiation	RadAvg	1.90	2.95	3.85	4.86	5.59	6.26	6.66	5.50	4.38	2.96	2.03	1.51	
	RadStd	0.17	0.23	0.31	0.45	0.41	0.41	0.24	0.42	0.33	0.35	0.14	0.12	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (38 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page